

A Countably Additive Dilation for Operator-Valued Quantum Measures

Run fa_banach_001, agent_lane_06

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Source and Open Question

The source paper is Han–Hu–Larson–Liu, *Dilations for operator-valued quantum measures*, arXiv:2109.09042 [2]. Its abstract states that a general projection-valued dilation is known for bounded quantum measures on projection lattices of von Neumann algebras with no type I_2 direct summand, but that such a dilation need not preserve countable additivity. The extracted open question is:

Does every countably additive $B(X)$ -valued quantum measure on the projection lattice of a von Neumann algebra with no type I_2 direct summand admit a countably additive projection-valued dilation?

The source paper proves this under extra hypotheses, namely $X = \ell_p$ for $1 \leq p < 2$, X has the Schur property, or the measure has bounded p -variation. The result below removes those extra hypotheses.

DILATIONS FOR OPERATOR-VALUED QUANTUM MEASURES

DEGUANG HAN, QIANFENG HU, DAVID R. LARSON, AND RUI LIU

ABSTRACT. This paper concerns the dilations of Banach space operator-valued quantum measures. While the recently developed general dilation theory can lead to a projection (idempotent) valued dilation for any quantum measure over the projection lattice for a von Neumann algebra that does not contain type I_2 direct summand, such a dilation does not necessarily guarantee the preservation of countable additivity of the quantum measure. So it remains an open question whether every countably additive $B(X)$ -valued quantum measure can be dilated to a countably additive projection-valued measure. The main purpose of this paper is to prove that such a dilation can be constructed if one of the following two conditions is satisfied: (i) the underlying Banach space $X = \ell_p$ ($1 \leq p < 2$) or it has Schur property, (ii) the quantum measure has bounded p -variation for some $1 \leq p < \infty$. All of these were achieved by establishing a non-commutative version of a minimal dilation theory on the so-called elementary dilation space equipping with an appropriate dilation norm. In particular, the newly introduced p -variation norm on the elementary dilation space allows us to prove that every operator-valued quantum measure with bounded p -variation has a projection-valued quantum measure dilation that preserves the boundedness of the p -variation.

Figure 1: The open-question sentence in the abstract of arXiv:2109.09042.

Proof Intuition

The obstruction in the elementary dilation norm of [2] is uniformity: after cutting by a tail projection $q_n \downarrow 0$, one has to control $\overline{U}(aq_n)x$ uniformly in $\|a\| \leq 1$. Rather than force this in the original Banach space X , place the dilation vectors in the von Neumann predual M_* . For each $x \in X$, the normal map $a \mapsto \overline{U}(a)x$ gives a weakly compact operator

$$Tx : X^* \longrightarrow M_*.$$

Right multiplication by projections on M_* is automatically countably additive on weakly compact subsets, by the Akemann–Jarchow weak compactness criterion. This supplies exactly the missing uniform tail estimate, while evaluation at the identity compresses the dilation back to X .

Statement

Theorem 1. *Let M be a sigma-finite von Neumann algebra with no direct summand of type I_2 , let X be a Banach space, and let*

$$U : \mathcal{P}(M) \longrightarrow B(X)$$

be a bounded countably additive operator-valued quantum measure. Then there are a Banach space Z , bounded operators $T : X \rightarrow Z$ and $S : Z \rightarrow X$, and a countably additive projection-valued quantum measure

$$V : \mathcal{P}(M) \longrightarrow B(Z)$$

such that

$$U(p) = SV(p)T \quad (p \in \mathcal{P}(M)).$$

In fact, the dilation extends to a normal bounded homomorphism $\pi : M \rightarrow B(Z)$ satisfying $\overline{U}(a) = S\pi(a)T$ for every $a \in M$, where $\overline{U} : M \rightarrow B(X)$ is the normal Bunce–Wright extension of U .

External Lemma Used

We use the following standard form of the Akemann–Jarchow weak compactness criterion for von Neumann preduals; see Akemann [1], Jarchow [4], and the modern JBW*-triple formulation in Hamhalter–Kalenda–Peralta–Pfitzner [3, Theorem 11.3].

Lemma 1 (uniform one-sided absolute continuity). *Let M be a von Neumann algebra and let $K \subset M_*$ be relatively weakly compact. If (q_n) is a decreasing sequence of projections in M with $q_n \downarrow 0$ strongly, then*

$$\sup_{\rho \in K} \|\rho \cdot q_n\| \longrightarrow 0, \quad \sup_{\rho \in K} \|q_n \cdot \rho\| \longrightarrow 0,$$

where $(\rho \cdot q)(a) = \rho(aq)$ and $(q \cdot \rho)(a) = \rho(qa)$.

Proof sketch of Lemma 1. One form of the Akemann–Jarchow criterion says that for a relatively weakly compact $K \subset M_*$ and every $\varepsilon > 0$, there is a positive normal functional $\varphi \in M_*^+$ and $\delta > 0$ such that

$$\|a\| \leq 1, \quad \varphi(a^*a) < \delta \implies |\rho(a)| < \varepsilon \quad (\rho \in K).$$

Since $q_n \downarrow 0$, normality gives $\varphi(q_n) \rightarrow 0$. For $\|b\| \leq 1$,

$$\varphi((bq_n)^*(bq_n)) = \varphi(q_nb^*bq_n) \leq \varphi(q_n),$$

so $\sup_{\rho \in K, \|b\| \leq 1} |\rho(bq_n)| \rightarrow 0$, which is exactly $\sup_{\rho \in K} \|\rho \cdot q_n\| \rightarrow 0$. The left-hand version follows by applying the same argument in the opposite von Neumann algebra. $\square$

Proof of the Theorem

By the Bunce–Wright extension theorem as used in [2], U extends uniquely to a bounded linear map

$$\overline{U} : M \longrightarrow B(X).$$

Because U is countably additive and M is sigma-finite, the extension is normal in the weak-operator sense: for every $x \in X$ and $x^* \in X^*$, the functional

$$a \longmapsto x^*(\overline{U}(a)x)$$

belongs to M_* .

Let $\mathcal{W} = \mathcal{W}(X^*, M_*)$ denote the Banach space of weakly compact operators from X^* to M_* , with the usual operator norm. For $a \in M$, define $\pi(a) : \mathcal{W} \rightarrow \mathcal{W}$ by right multiplication on M_* :

$$(\pi(a)A)(x^*) = A(x^*) \cdot a, \quad ((\rho \cdot a)(b) = \rho(ba)).$$

Then $\|\pi(a)\| \leq \|a\|$, and $\pi(ab) = \pi(a)\pi(b)$. Hence $\pi : M \rightarrow B(\mathcal{W})$ is a bounded homomorphism. In particular, for $p \in \mathcal{P}(M)$, $\pi(p)$ is an idempotent and $\pi(p+q) = \pi(p) + \pi(q)$ whenever $p \perp q$.

For $x \in X$, define $T_0x : X^* \rightarrow M_*$ by

$$(T_0x)(x^*)(a) = x^*(\overline{U}(a)x), \quad a \in M.$$

This is bounded with $\|T_0x\| \leq \|\overline{U}\| \|x\|$. It is weakly compact: the unit ball of X^* , with its weak-star topology, is compact, and the map $x^* \mapsto (T_0x)(x^*)$ is weak-star to weak continuous because evaluation at each $a \in M$ is the scalar map

$$x^* \longmapsto x^*(\overline{U}(a)x).$$

Thus $T_0x \in \mathcal{W}$.

Let

$$Z = \overline{\text{span}}\{\pi(a)T_0x : a \in M, x \in X\} \subseteq \mathcal{W},$$

and let $T : X \rightarrow Z$ be $T = T_0$. Define S first on the algebraic span by

$$S \left(\sum_{j=1}^m \pi(a_j)T x_j \right) = \sum_{j=1}^m \overline{U}(a_j)x_j.$$

This is well-defined. Indeed, if the vector inside Z is zero, then for every $x^* \in X^*$,

$$0 = \left(\sum_j \pi(a_j)T x_j \right) (x^*)(1) = x^* \left(\sum_j \overline{U}(a_j)x_j \right),$$

so the displayed vector in X is zero. Moreover,

$$\begin{aligned} \left\| S \left(\sum_j \pi(a_j)T x_j \right) \right\| &= \sup_{\|x^*\| \leq 1} \left| \left(\sum_j \pi(a_j)T x_j \right) (x^*)(1) \right| \\ &\leq \left\| \sum_j \pi(a_j)T x_j \right\|_{\mathcal{W}}. \end{aligned}$$

Therefore S extends to a contraction $S : Z \rightarrow X$.

For every $a \in M$ and $x \in X$,

$$S\pi(a)Tx = \overline{U}(a)x.$$

Indeed, applying $x^* \in X^*$ to the left side gives

$$(\pi(a)Tx)(x^*)(1) = (Tx)(x^*)(a) = x^*(\overline{U}(a)x).$$

It remains to check countable additivity on projections. Let $(p_j)_{j \geq 1}$ be mutually orthogonal projections, and put

$$p = \sum_{j=1}^{\infty} p_j, \quad q_n = p - \sum_{j=1}^n p_j.$$

Then $q_n \downarrow 0$ strongly. For any $A \in \mathcal{W}$, the set $K = A(\text{Ball}_{X^*})$ is relatively weakly compact in M_* , so Lemma 1 yields

$$\|\pi(q_n)A\| = \sup_{\|x^*\| \leq 1} \|A(x^*) \cdot q_n\| = \sup_{\rho \in K} \|\rho \cdot q_n\| \longrightarrow 0.$$

Consequently $\pi(p)A = \sum_{j=1}^{\infty} \pi(p_j)A$ in norm for every $A \in Z$. Thus

$$V(p) = \pi(p)|_Z, \quad p \in \mathcal{P}(M),$$

is a countably additive projection-valued quantum measure on Z . The compression identity $U(p) = SV(p)T$ follows from the identity already proved for all $a \in M$.

Finally, the same estimate shows that π is normal in the strong operator sense relevant to the projection lattice: decreasing projection tails act strongly as zero on Z . This is stronger than what is needed for the stated dilation problem. $\square$

Verification Notes

No computational search was used. The proof reduces the source problem to a standard weak compactness criterion for subsets of M_* . The key checks for a human reviewer are:

1. $T_0x : X^* \rightarrow M_*$ is weakly compact because it is weak-star to weak continuous on the weak-star compact unit ball of X^* .
2. The compression S is well-defined and contractive because evaluation at the identity is bounded on M_* .
3. Countable additivity is exactly Lemma 1 applied to the weakly compact set $A(\text{Ball}_{X^*})$.

Novelty Check and Limitations

I searched the run indexes for arXiv:2109.09042 and for the phrases “operator-valued quantum measure”, “projection-valued quantum measure”, “countably additive”, “Naimark”, “Stinespring”, and “dilation”. No existing packet or attempt for this source paper was present. I also performed a bounded web search for exact phrases from the question and for later papers on countably additive projection-valued dilations of operator-valued quantum measures. I found the source paper and background weak compactness literature, but no later paper explicitly resolving this exact question.

The result is therefore recorded as a candidate full solution, subject to human review. The main review focus should be the exact form of the Akemann–Jarchow criterion in Lemma 1; once that standard lemma is accepted, the dilation construction is direct.

References

- [1] C. A. Akemann, *The dual space of an operator algebra*, Trans. Amer. Math. Soc. 126 (1967), 286–302.
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- [3] J. Hamhalter, O. F. K. Kalenda, A. M. Peralta, and H. Pfizner, *Measures of weak non-compactness in preduals of von Neumann algebras and JBW^* -triples*, arXiv:1901.08056, 2019.
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